BEGINNER-FRIENDLY · EXAM-READY · COMPLETE NOTES

Machine Learning

A clear, structured guide to the core concepts, mathematical intuition, worked examples, and professional answers you can adapt in an exam.

14 core topics | Formulas explained | Exam-answer boxes | Revision sheet

Understand first

Plain-language intuition comes before formal definitions and equations.

Write confidently

Exam-ready responses include the key definition, mechanism, use case and limitations.

Revise efficiently

Comparison tables, formula summaries and common exam traps are collected at the end.

READ

How to use this handbook

For every topic, first read the intuition, then understand the formula, and finally practise writing the exam answer without looking. For a 5-mark response, aim for a precise definition, the central mechanism or equation, one example, and relevant advantages or limitations.

Notation: $P(A \mid B)$ means the probability of event A given that B is observed. In regression, x denotes input features and y denotes the target. If your instructor uses different symbols, the underlying ideas remain the same.

01

Bayesian Learning and Bayes' Theorem

Beginner intuition

Bayesian learning updates what we believe about a hypothesis when new evidence arrives. Rather than treating a hypothesis as simply true or false, it represents uncertainty with probabilities. As training examples are observed, the probability assigned to each hypothesis changes.

Bayes' theorem

$$P(H \mid D) = \frac{P(D \mid H)\, P(H)}{P(D)}$$

Component	Meaning in machine learning	Example
Prior, $P(H)$	Belief in hypothesis H before seeing the current data.	Before reviewing an email, assume a 10% spam rate.
Likelihood, $P(D \mid H)$	Probability of observing data D if H were true.	How likely is the word "prize" if the email is spam?
Evidence, $P(D)$	Overall probability of observing D under the hypotheses considered.	How common is this word across all emails?
Posterior, $P(H \mid D)$	Updated probability of H after observing D.	Revised probability that the email is spam.

For two mutually exclusive hypotheses H and $\neg H$, the evidence is:

$$P(D) = P(D \mid H)P(H) + P(D \mid \neg H)P(\neg H)$$

This normalises the posterior probabilities so that they sum to 1.

Bayesian learning: general procedure

1. Specify hypotheses or model parameters and their prior probabilities.
2. Observe training data and evaluate how likely it is under each hypothesis.
3. Apply Bayes' theorem to obtain posterior probabilities.
4. Predict using the posterior distribution, or select the most probable hypothesis if required.

EXAM-READY ANSWER

Bayes' theorem updates the probability of a hypothesis after observing data. The posterior equals the likelihood multiplied by the prior, divided by the evidence. The prior represents belief before observing the data; the likelihood measures how well the hypothesis explains the data; the evidence normalises the result; and the posterior is the updated belief. Bayesian learning is used in probabilistic classification, parameter estimation and reasoning under uncertainty.

02

Bayes Optimal Classifier

Core idea

The Bayes Optimal Classifier considers every possible hypothesis, weights each hypothesis by its posterior probability, and combines their predictions. It chooses the class with the greatest posterior-weighted support. Under the true data distribution and ordinary 0–1 classification loss, it minimises expected classification error.

Mathematical form

$$\hat{y} = \arg\max_{c} \sum_{h} P(c \mid x, h)\, P(h \mid D)$$

Here x is a new example, h is a possible hypothesis, D is the training data, and $P(h \mid D)$ is the posterior probability of that hypothesis. For deterministic hypotheses, $P(c \mid x, h)$ is 1 if h predicts class c, and 0 otherwise.

Example

Suppose hypotheses give total posterior-weighted support of 0.90 to “spam” and 0.10 to “not spam.” The Bayes Optimal prediction is spam.

Strengths and limitation

- Uses uncertainty across hypotheses instead of relying on a single selected hypothesis.
- Minimises expected classification error under the assumed distribution and 0–1 loss.
- Usually expensive or infeasible to compute exactly because the hypothesis space may be huge.

EXAM-READY ANSWER

The Bayes Optimal Classifier predicts the class with the highest posterior-weighted probability across all hypotheses. It combines predictions according to each hypothesis’s posterior probability instead of choosing only one hypothesis. Under the true distribution and 0–1 loss, it minimises expected classification error. Its main limitation is computational cost.

03

Naïve Bayes Classifier

Definition and intuition

Naïve Bayes is a probabilistic classifier based on Bayes' theorem. It assumes that features are conditionally independent of one another given the class label. This simplifying assumption is often imperfect, but it makes learning and prediction efficient.

Classification rule

$$P(C \mid x_1, \ldots, x_n) \propto P(C) \prod_{i=1}^{n} P(x_i \mid C)$$

Calculate this score for each class and select the class with the largest score. Normalisation is needed if actual posterior probabilities are required.

What does the naïve assumption mean?

Given that an email is spam, the model treats the occurrence of "prize" as independent of the occurrence of "winner." Real words can be related; Naïve Bayes deliberately ignores such dependencies. Independence is assumed conditional on the class.

Computational complexity

For N training examples, d features and K classes, training is typically $O(Nd)$ for counting or estimating feature statistics. Prediction for one example is typically $O(Kd)$. Storage for a simple categorical model is roughly $O(Kd)$, depending on vocabulary and representation.

Zero-frequency problem

If a feature value never occurs with a class in the training data, its estimated likelihood can be zero and make the entire class score zero. Laplace smoothing adds a small count, commonly 1, to possible outcomes so unseen values do not automatically eliminate a class.

EXAM-READY ANSWER

Naïve Bayes is a supervised probabilistic classifier that applies Bayes' theorem with the assumption that features are conditionally independent given the class. It selects the class maximising $P(C) \prod_i P(x_i \mid C)$. This assumption reduces the number of probabilities to estimate, giving efficient training (typically $O(Nd)$) and prediction (typically $O(Kd)$). It is useful for text classification and spam filtering, but correlated features can reduce accuracy. Laplace smoothing handles unseen feature values.

04

Linear Regression vs Logistic Regression

The key distinction

Linear regression predicts a numerical quantity, such as house price or temperature. Logistic regression predicts a class probability, usually for binary classification. Despite its name, logistic regression is primarily a classification method.

Aspect	Linear Regression	Logistic Regression
Output	Continuous numeric value.	Probability between 0 and 1; class assigned using a threshold.
Model	$\hat{y} = \beta_0 + \sum_i \beta_i x_i$	$z = \beta_0 + \sum_i \beta_i x_i$; $p = 1/(1 + e^{-z})$
Use cases	Price, sales, demand or temperature prediction.	Spam detection, churn prediction, default prediction.
Common objective	Minimise squared error.	Minimise log loss / maximise likelihood.
Interpretation	Coefficient describes change in predicted target per feature unit.	Coefficient changes the log-odds of the positive class.
Output range	Any real number.	Probability lies between 0 and 1.

Logistic regression

$$z = \beta_0 + \sum_i \beta_i x_i \qquad p = \sigma(z) = \frac{1}{1 + e^{-z}}$$

The sigmoid function maps a real-valued score to a probability. At a threshold of 0.5, predict class 1 when $p \geq 0.5$ and class 0 otherwise. The threshold can be adjusted to reflect the costs of false positives and false negatives.

EXAM-READY ANSWER

Linear regression predicts continuous numerical outputs by fitting a linear relationship between features and the target. Logistic regression estimates the probability of a class by applying the sigmoid function to a linear score. Linear regression commonly minimises squared error, whereas logistic regression commonly minimises log loss. House-price prediction is a linear-regression task; predicting customer default is a logistic-regression task.

05

Case-Based Reasoning (CBR)

Definition

Case-Based Reasoning solves a new problem by retrieving and adapting solutions from similar past problems, rather than deriving every solution from general rules alone.

The 4R cycle

Stage	What happens	Example: laptop troubleshooting
Retrieve	Find a past case similar to the current problem.	Find cases with overheating and sudden shutdown.
Reuse	Apply or adapt the old solution.	Try a previously successful fan-cleaning procedure.
Revise	Test the solution and correct it if needed.	Check temperatures; investigate further if it still shuts down.
Retain	Store the new case and its outcome.	Save the confirmed cause and solution for future use.

Key components

- **Case base:** stored problems, contexts, solutions and outcomes.
- **Similarity measure:** finds the closest past cases.
- **Adaptation method:** modifies a retrieved solution for the new case.
- **Evaluation and learning:** tests the solution and stores useful experience.

Strengths and limitations

CBR reuses experience, can justify recommendations through precedent, and improves as useful cases accumulate. However, retrieval depends on case quality and similarity design; old solutions may not transfer, and a case library can become large or inconsistent.

EXAM-READY ANSWER

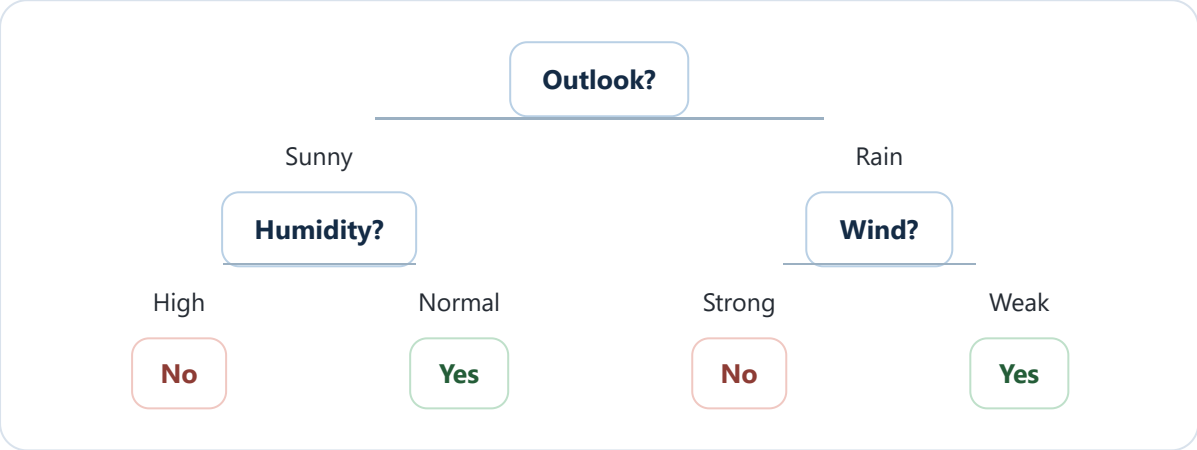
Case-Based Reasoning solves new problems using experiences from similar past cases. It follows the 4R cycle: Retrieve a similar case, Reuse its solution, Revise it after testing, and Retain the new experience. CBR is useful in diagnosis, customer support and technical decision support. Its performance depends on the quality of the case library, similarity measure and adaptation process.

06

Decision Trees: Concept, Construction and Issues

What is a decision tree?

A decision tree is a supervised-learning model that represents decisions as a tree. Internal nodes test features, branches represent test outcomes, and leaves contain the final class prediction or numerical estimate. Prediction follows a path from the root to a leaf.



How to read it: start at the root, test the feature, follow the matching branch, and continue until reaching a leaf that gives the prediction.

How it works

1. Start with all training examples at the root.
2. Choose a feature test that improves class purity or reduces prediction error.
3. Split the examples into child nodes and repeat recursively.
4. Stop when a stopping condition is met and assign a prediction to each leaf.

Types and uses

Classification trees predict categories such as approve/reject. Regression trees predict numerical values, often using the mean target in a leaf. Applications include risk scoring, fault diagnosis and decision support.

Advantages

- Easy to visualise and explain; little feature scaling is required.
- Can model nonlinear relationships and feature interactions.
- Can handle numerical and categorical features, depending on implementation.

Issues / disadvantages

- **Overfitting:** a deep tree can memorise noise or individual examples.

- **Instability:** small changes in training data may produce a different tree.
- **Greedy splitting:** locally best choices do not guarantee a globally optimal tree.
- Some split criteria favour attributes with many distinct values.
- Large trees are difficult to interpret and may create tiny leaves.
- Axis-aligned splits may require many nodes for some decision boundaries.
- Class imbalance, noise and missing values can distort split decisions.

EXAM-READY ANSWER

A decision tree is a supervised model containing internal decision nodes, branches and leaf nodes. It recursively partitions data using features that improve class purity or reduce prediction error. Trees are interpretable and model nonlinear patterns, but may overfit, be unstable, make greedy local choices, and become difficult to interpret when large. Depth limits, minimum leaf sizes, pruning and ensemble methods can mitigate these problems.

07

Overfitting in Decision Trees

Meaning and symptoms

Overfitting occurs when a model learns accidental details and noise in the training set instead of the underlying pattern. A very deep tree may classify nearly every training example correctly but perform poorly on unseen data.

- Training accuracy is very high, but validation/test accuracy is much lower.
- The tree has many levels and leaves containing very few examples.
- Small changes to the training sample produce a substantially different tree.

Why it happens

A tree can create narrow rules for individual examples. Additional splits may improve training fit but increase variance and reduce generalisation, especially with noisy or limited data.

Prevention methods

Method	How it helps
Pre-pruning	Stop growth early using maximum depth, minimum samples per split/leaf, or minimum impurity decrease.
Post-pruning	Grow a larger tree, then remove branches that do not improve validation performance enough.
Cross-validation	Compare settings on held-out folds rather than training accuracy alone.
Representative data	Reduces the chance that random noise appears to be a stable rule.
Ensembles	Random forests and boosting can improve prediction, usually with less simple interpretability.

EXAM-READY ANSWER

Overfitting in decision trees occurs when the tree becomes too complex and captures noise or peculiarities of the training data. It often yields high training accuracy but poorer performance on unseen data. It can be reduced using pre-pruning, such as limiting depth and minimum leaf size, or post-pruning, which removes branches that do not generalise. Cross-validation helps select complexity.

08

Entropy in Binary Classification

Intuition

Entropy measures uncertainty or impurity in a set of labels. If every example belongs to one class, entropy is zero. If positive and negative examples are equally common, entropy is maximal for a binary label.

Binary entropy formula

$$H(Y) = -p\log_2(p) - (1-p)\log_2(1-p)$$

Here p is the proportion of positive examples and $1-p$ is the proportion of negative examples. By convention, $0\log_2(0)$ is treated as 0. With base-2 logarithms, entropy is measured in bits.

Examples

Positive proportion p	Negative proportion $1-p$	Entropy	Interpretation
1.0	0.0	0 bits	Completely pure
0.75	0.25	≈ 0.811 bits	Some uncertainty
0.5	0.5	1 bit	Maximum binary uncertainty
0.0	1.0	0 bits	Completely pure

Worked calculation for $p = 0.75$

$$H = -0.75\log_2(0.75) - 0.25\log_2(0.25)$$

$$\approx -0.75(-0.415) - 0.25(-2) = 0.811 \text{ bits}$$

Entropy and information gain

A decision-tree split is useful when the child subsets have lower weighted entropy than the parent.

$$\text{Gain}(S, A) = H(S) - \sum_{v} \frac{|S_v|}{|S|} H(S_v)$$

Here S is the current dataset, A is the candidate attribute, and S_v is the subset with attribute value v. ID3 chooses the attribute with the greatest information gain.

EXAM-READY ANSWER

For binary classification, entropy quantifies uncertainty in the class label. If p is the proportion of positive cases, $H(Y) = -p \log_2 p - (1-p) \log_2 (1-p)$. Entropy is zero when all cases belong to one class and reaches 1 bit when both classes occur with probability 0.5. Information gain measures the reduction in entropy after a split.

09

ID3 Algorithm and Its Inductive Bias

What is ID3?

Iterative Dichotomiser 3 (ID3) is a top-down decision-tree algorithm. At each node, it selects the attribute with the highest information gain, splits the examples by that attribute, and recursively repeats on each subset.

Algorithm

1. If all examples at a node have the same class, return a leaf with that class.
2. If no attributes remain, return a majority-class leaf.
3. Calculate information gain for every available attribute.
4. Select the attribute with the greatest information gain.
5. Split examples by its values and recursively build each child subtree.

Compact pseudocode

$$a^* = \arg\max_{a \in A} \text{Gain}(S, a)$$

Create a node testing a^*. For each value v, recursively build a subtree from examples in S where $a^* = v$, using the remaining attributes. Stop when the examples are pure or no attributes remain.

Inductive bias regarding tree size

ID3 favours attributes that reduce entropy strongly and early. This can produce compact trees when an informative feature creates a large reduction near the root. However, basic ID3 does not explicitly minimise the number of nodes or guarantee the smallest possible tree. It makes greedy local choices and can still produce a large, overfit tree.

Limitations

- Basic ID3 is mainly designed for categorical features; continuous features need discretisation or an extension.
- Information gain can favour attributes with many distinct values.
- Basic ID3 does not inherently perform robust pruning.
- Greedy local choices do not guarantee a globally optimal tree.

EXAM-READY ANSWER

ID3 is a greedy, top-down decision-tree algorithm that selects the attribute with maximum information gain at each node. Its inductive bias favours attributes that reduce entropy strongly and early, often leading to compact trees when informative attributes exist. However, it does not explicitly optimise tree size or guarantee the minimum-size tree, and it can overfit. Information gain may also favour attributes with many values.

10

k-Nearest Neighbours and Lazy Learning

Definition

k-Nearest Neighbours (k-NN) predicts a new example using the k closest stored training examples. Classification commonly uses majority voting; regression commonly averages neighbours' target values.

Why is it called lazy learning?

It is called lazy learning because it performs little explicit model-building during training. It stores the examples and postpones most computation until a query arrives. Eager learners, by contrast, build a model before prediction.

How it works

1. Choose k.
2. Scale features when their numeric ranges differ substantially.
3. Calculate distances from the query to training examples.
4. Select the k closest examples and vote or average their target values.

Euclidean distance

$$d(x, z) = \sqrt{\sum_i (x_i - z_i)^2}$$

Trade-offs

Advantages	Limitations
Simple and intuitive; no strong parametric form is assumed.	Prediction may be slow because many examples need distance calculations.
Can model complex decision boundaries.	Sensitive to scaling, irrelevant features and distance choice.
Easy to update by adding examples.	Memory-intensive; distance becomes less useful in high dimensions.

Complexity

With N examples and d features, a straightforward query costs $O(Nd)$ to compute distances, plus neighbour selection. Training is minimal, but storing the dataset costs $O(Nd)$.

EXAM-READY ANSWER

k-NN predicts a query by finding its k nearest stored examples and using majority voting for classification or averaging for regression. It is called lazy learning because it builds little explicit model during training and delays computation until prediction. It is simple and flexible but can be slow at inference, memory-intensive and sensitive to feature scaling.

11

Support Vector Machines (SVM)

Core idea

An SVM is a supervised-learning method that finds a decision boundary separating classes. A linear SVM chooses the hyperplane with the largest margin. Kernel functions allow nonlinear decision boundaries without explicitly constructing all high-dimensional features.

Key terms

Term	Meaning
Hyperplane / decision surface	The boundary separating classes. A linear hyperplane has equation $w^\top x + b = 0$.
Support vectors	Training points closest to the boundary; they determine the maximum-margin solution.
Margin	The distance from the boundary to the nearest training points. The canonical margin width is $2/\|w\|$.
Kernel	A similarity function that acts as if data were mapped to a richer feature space; examples include linear, polynomial and RBF kernels.

Decision surface and margin

$$\text{Hyperplane: } w^\top x + b = 0$$

$$\text{Margin boundaries: } w^\top x + b = +1 \quad \text{and} \quad w^\top x + b = -1$$

$$\text{Margin width} = \frac{2}{\|w\|}$$

Maximising the margin is equivalent to minimising $\|w\|^2/2$ under the canonical constraints. The support vectors are the points that lie on or inside the margin boundaries.

Hard margin vs soft margin

- Hard-margin SVM assumes perfectly linearly separable training data.
- Soft-margin SVM uses slack variables to allow margin violations or misclassification; C controls the trade-off between margin size and violations.
- Kernel SVMs can learn nonlinear boundaries, but kernel and parameter selection matter.

Advantages and limitations

SVMs can perform well in high-dimensional spaces and use kernels for nonlinear patterns. However, parameter selection can be important, probability estimates need calibration, large kernel problems can be expensive, and a model may be less interpretable than a small tree.

EXAM-READY ANSWER

A Support Vector Machine constructs a decision surface separating classes, usually by maximising the margin. Its hyperplane is $w^\top x + b = 0$. Support vectors are the closest training points that determine the boundary, while the margin measures separation from those points. Soft-margin SVMs allow violations, and kernel functions enable nonlinear classification. SVMs can work well in high-dimensional spaces but require careful parameter selection.

12

Expectation-Maximization (EM) Algorithm

Why EM is needed

Some statistical models contain hidden or missing variables. Direct likelihood maximisation can be difficult because hidden assignments are unknown. EM alternates between estimating hidden-variable distributions and updating model parameters.

The two steps

Step	Operation	Plain-language meaning
E-step (Expectation)	Use current parameters to calculate the posterior distribution of latent variables and the expected complete-data log-likelihood.	Estimate likely hidden explanations.
M-step (Maximisation)	Update parameters to maximise the expectation calculated in the E-step.	Adjust the model to fit those explanations.

Algorithm

1. Initialise parameters θ.
2. E-step: compute $Q(\theta \mid \theta^{(t)}) = \mathbb{E}[\log p(X, Z \mid \theta) \mid X, \theta^{(t)}]$.
3. M-step: choose $\theta^{(t+1)}$ that maximises $Q(\theta \mid \theta^{(t)})$.
4. Repeat until the likelihood or parameter changes become sufficiently small.

Example: Gaussian mixture model

Suppose points come from two overlapping clusters, but cluster membership is unknown. The E-step estimates each point's probability of belonging to each Gaussian component. The M-step updates each component's mean, variance and mixture weight using those soft assignments.

Properties

- Under standard conditions, EM does not decrease observed-data likelihood at each iteration.
- It typically converges to a local optimum or stationary point, not necessarily a global optimum.
- Results depend on initialisation; multiple starts can help.
- Convergence can be slow near a solution.

EXAM-READY ANSWER

Expectation-Maximization is an iterative method for estimating parameters when data contains latent or missing variables. The E-step estimates the distribution of hidden variables or expected complete-data log-likelihood using current parameters. The M-step maximises this expectation to update parameters. The steps repeat until convergence. EM is used in Gaussian mixture models and missing-data estimation; it generally preserves or improves likelihood per iteration but may converge to a local optimum.

13

Radial Basis Function (RBF) Networks

Definition

An RBF network is a feed-forward neural network whose hidden units respond to the distance between an input and a centre. The output layer combines these responses to produce a regression value or class score.

Architecture

Layer	Role
Input layer	Receives feature vector x.
RBF hidden layer	Measures closeness to centres c_j, commonly with Gaussian functions.
Output layer	Combines hidden activations, often using a weighted linear sum.

Gaussian radial basis function

$$\phi_j(x) = \exp\left(-\frac{\|x - c_j\|^2}{2\sigma_j^2}\right)$$

Here c_j is the centre of hidden unit j, σ_j controls its width, and $\|x - c_j\|$ is the distance to the centre. Inputs near a centre produce activation near 1; distant inputs produce activation near 0.

How it is trained

1. Choose centres, for example using k-means clustering or selected training examples.
2. Set widths using distances among centres or another suitable rule.
3. Fit output-layer weights, often using least squares for regression or a classification objective.

Advantages and limitations

RBF networks offer local responses, can approximate nonlinear functions and can have a straightforward output-layer fit once centres and widths are fixed. Their performance depends on centre and width selection; too many units increase cost, and high-dimensional data can be challenging.

EXAM-READY ANSWER

An RBF network is a feed-forward neural network with input, radial-function hidden and output layers. Each hidden unit responds to distance from a centre, often using $\phi(x) = \exp(-\|x - c\|^2/(2\sigma^2))$. The output layer combines these activations. RBF networks are useful for nonlinear classification and function approximation, but performance depends on centre and width selection.

14 Quick Revision: Comparisons and Formula Sheet

High-yield comparisons

Concept	Remember this distinction
Bayes Optimal vs Naïve Bayes	Bayes Optimal combines predictions across hypotheses; Naïve Bayes uses conditional feature independence.
Linear vs Logistic Regression	Continuous numerical output vs class probability.
Eager vs Lazy Learning	Eager learners build a model during training; k-NN stores examples and computes at query time.
Entropy vs Information Gain	Entropy measures uncertainty; information gain measures uncertainty reduction after a split.
Pre-pruning vs Post-pruning	Pre-pruning stops growth early; post-pruning removes branches after growth.
SVM margin vs support vectors	Margin is the separation distance; support vectors are the closest points that define it.
EM E-step vs M-step	E-step estimates hidden-variable distributions; M-step updates parameters.
RBF network vs RBF-kernel SVM	RBF network uses radial hidden units; an RBF-kernel SVM uses a kernel similarity in SVM optimisation.

Formula sheet

Topic	Formula / rule
Bayes' theorem	$P(H \mid D) = P(D \mid H)P(H)/P(D)$
Binary entropy	$H(p) = -p\log_2 p - (1-p)\log_2(1-p)$
Information gain	$\mathrm{Gain}(S, A) = H(S) - \sum_v (\lvert S_v\rvert/\lvert S\rvert)H(S_v)$
Logistic regression	$p = 1/(1+e^{-z}), \quad z = \beta_0 + \sum_i \beta_i x_i$
Euclidean distance	$d(x, z) = \sqrt{\sum_i (x_i - z_i)^2}$
SVM hyperplane	$w^\top x + b = 0$; canonical margin width $= 2/\lVert w\rVert$
Gaussian RBF	$\phi_j(x) = \exp(-\lVert x - c_j\rVert^2/(2\sigma_j^2))$
Naïve Bayes score	$P(C)\prod_i P(x_i \mid C)$, then choose the largest class score

How to write a strong 5-mark answer

1. Start with a precise definition.
2. Explain the principle or mechanism in ordered steps.

3. Include the central equation or a labelled diagram and define its symbols.
4. Give one realistic example or use case.
5. State a relevant advantage and limitation when appropriate.

Common exam traps

- Logistic regression estimates class probabilities; it does not predict an unbounded continuous target.
- ID3 does not guarantee the smallest tree; it makes greedy information-gain choices.
- Entropy measures uncertainty; information gain measures uncertainty reduction.
- k-NN stores training data; it does not mean there is no training dataset.
- EM does not guarantee a global optimum.
- An RBF network is not the same model as an SVM using an RBF kernel.

FINAL MEMORY MAP

Bayes updates beliefs; Naïve Bayes classifies using an independence shortcut; regression predicts numbers or class probabilities; CBR reuses past cases; trees split data; entropy measures uncertainty; ID3 greedily maximises information gain; k-NN delays computation until prediction; SVM maximises margin; EM alternates expectation and maximisation; RBF networks use distance-based hidden activations.

End of handbook

Use the exam answers as structures to practise, not as text to memorise blindly. Understanding the intuition makes it easier to answer differently worded questions.